

Plant nitrogen uptake preference and drivers in natural ecosystems at the global scale

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Summary

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Received: 27 November 2024

Accepted: 27 January 2025

New Phytologist (2025) **246**: 972–983

doi: 10.1111/nph.70030

Key words: ammonium, nitrate, nitrogen deposition, nitrogen uptake preference, organic nitrogen, stable isotopes.

- Elucidating plant nitrogen (N) acquisition is crucial for understanding plant N strategies and ecosystem productivity. However, the variation in plant N uptake preference and its controlling factors on a global scale remain unclear.
- We conducted a global synthesis to explore plant N preference patterns and driving factors.
- Globally, the average contributions of ammonium (NH_4^+), nitrate (NO_3^-), and glycine N to the total plant N uptake were $41.6 \pm 1.1\%$, $32.8 \pm 1.2\%$, and $25.6 \pm 0.9\%$, respectively. However, plant N uptake preferences differed significantly among climatic regions and vegetation types. Soil NH_4^+ was the most preferred N form by plants in (sub)tropical regions, whereas NO_3^- preference was significantly higher in high-latitude than low-latitude regions. Plant functional type was one of the most important factors driving NO_3^- preference, with significantly higher NO_3^- preference of nonwoody species than broadleaf-evergreen, conifer, and shrub species. Organic N preference was lowest in (sub)tropics and significantly lower than that in temperate and alpine regions.
- This study shows clear climatic patterns and different influencing factors of plant NH_4^+ and NO_3^- preference, which can contribute to the accurate prediction of N constraints on ecosystem productivity and soil carbon dynamics.

Introduction

Nitrogen (N) is the primary nutrient element limiting the growth and productivity of terrestrial plants (Mason *et al.*, 2022). Plant preference for different soil N forms is one of the most critical strategies for acquiring soil N resources because it can mitigate N competition among plants, maintain plant growth, and contribute to ecosystem carbon (C) sequestration (McKane *et al.*, 2002). However, the relative importance of different soil N for plants is poorly understood. Between soil ammonium N (NH_4^+ -N) and nitrate N (NO_3^- -N), which have been considered the main sources of N for plants, plants may prefer NH_4^+ uptake to NO_3^- uptake due to its lower energy cost and faster assimilation (County *et al.*, 2015; Moreau *et al.*, 2019). However, NH_4^+ nutrition often induces a growth-limiting stress known as ammonium syndrome, which may offset the energy-saving advantage of NH_4^+ uptake over NO_3^- (Britto & Kronzucker, 2013). Accordingly, some plant species, such as *Ericaceae* or spruce, have been found to prefer NH_4^+ uptake (Pearson & Stewart, 1993; Britto & Kronzucker, 2013), whereas, many fast-growing annual grasses and some pioneer woody plants have been found to prefer NO_3^- uptake (Pearson & Stewart, 1993; Yi *et al.*, 2023). Moreover, NO_3^- is more mobile than NH_4^+ ; therefore, it can move more easily to the root surface, presumably

giving plants a better chance of N uptake (Zhou *et al.*, 2021). These physiological costs and physical limitations constrain plant strategies in opposite directions, leading to considerable uncertainty regarding the relative importance of soil NH_4^+ and NO_3^- uptake for different plant species. In addition to NH_4^+ and NO_3^- , plants have recently been shown to directly assimilate soil organic N (Näsholm *et al.*, 2009; Inselsbacher & Wanek, 2021). For instance, using ^{15}N labelled model amino acid, previous studies have shown direct organic N uptake by a variety of plant species in arctic (Nordin *et al.*, 2004), boreal (Persson & Näsholm, 2008; Näsholm *et al.*, 2009), alpine (Miller & Bowman, 2003; Xu *et al.*, 2006) and temperate ecosystems (Rains & Bledsoe, 2007). However, our knowledge of the global patterns of plant N uptake preferences for three N forms (NH_4^+ , NO_3^- , and organic N) remains limited.

The limited understanding of plant N uptake preference may be partly due to the complex interactions between plant N acquisition and multiple factors (Moreau *et al.*, 2019). Plant N acquisition strategies depend on various biotic and abiotic factors, such as plant functional types (Andersen *et al.*, 2017; Hill *et al.*, 2019), soil properties (Nordin *et al.*, 2004; Zhang *et al.*, 2016), and climatic conditions (Hill *et al.*, 2011; Boczulak *et al.*, 2014). Plant functional types represent the key functional strategies of plants in relation to their environment and provide

important insights into plant ecological responses and adaptations to highly variable habitats (Gillison, 2013). Although research on plant N uptake preference strategies has been conducted for various plant species and terrestrial ecosystems (Harrison *et al.*, 2008; Britto & Kronzucker, 2013; Liu *et al.*, 2017), general patterns of the N uptake preference of terrestrial plants at the species level remain unclear.

Soil N availability is considered a key factor influencing plant N uptake preferences, as it provides N resources from the soil to plants (Zhang *et al.*, 2021). It is plausible that dominant plant species have evolved unique N uptake strategies to maximize their N acquisition in different climate zones, with different soil N form availability (Mason *et al.*, 2022). In particular, in environments with severely limited soil N, such as alpine and boreal regions, there are notable differences in plant N uptake preferences for different soil N forms (NH_4^+ , NO_3^- or organic N) to mitigate strong competition for N (McKane *et al.*, 2002). However, this is not always the case. In fact, plant N uptake preferences for different soil N forms in natural ecosystems often vary between study sites regardless of N-rich (Liu *et al.*, 2017; Gurmesa *et al.*, 2019) or N-limited conditions (Liu *et al.*, 2018; Zhang *et al.*, 2021). Therefore, plant N uptake may be significantly influenced by factors other than soil N availability.

For instance, changes in temperature directly affect plant N acquisition (Warren, 2009; Boczulak *et al.*, 2014; Hu *et al.*, 2024) due to temperature-dependent active transport of soil N at the root–soil interface (Bassirirad, 2000). Temperature can also indirectly affect plant N preference by influencing microbial activity and N cycling rates (Dawes *et al.*, 2017; Lie *et al.*, 2020), which, in turn, can alter the availability of different soil N forms and competition for soil N between plants and microbes (Kuzyakov & Xu, 2013). However, few studies have investigated the effects of temperature on plant N preference. Soil water content is another important factor affecting plant N absorption. Previous theoretical and experimental studies have demonstrated that root N uptake is constrained by the rate of N transport toward the root surface (Oyewole *et al.*, 2016; McMurtrie & Näsholm, 2018). A high rate of transport toward the root surface enhances root N uptake by reducing the opportunity for soil microbes to intercept dissolved N and steepening the N concentration gradient in the rhizosphere adjacent to the root surface (McMurtrie & Näsholm, 2018; Henriksson *et al.*, 2021). However, as NO_3^- is more mobile than NH_4^+ in the soil solution, soil water content may differently influence plant N preference. Thus, multiple drivers of plant N uptake under different conditions should be identified to gain a better understanding of the relative importance of different forms of N to plants. Along a 3000 km transect spanning temperate and alpine grasslands with 18 species, a recent study found that, among multiple factors, soil N availability controls the uptake of different forms of N by plants (Liu *et al.*, 2024). However, on the global scale, with highly variable climatic conditions and species types, the factors driving plant N uptake preference remain unclear.

In this study, we compiled a global dataset of 101 publications on the N uptake strategies of different plant species in natural terrestrial ecosystems. The objectives of this study were (1) to

investigate the global pattern of plant preference for different forms of N and (2) to explore the main driving factors behind the different N uptake preferences of plants around the world. We hypothesized that, on a global scale, plant N uptake preferences for different soil N forms vary across climatic regions. Specifically, we hypothesized that (H1) soil NH_4^+ uptake is preferred under warm conditions (e.g. in the tropics), whereas organic N uptake is preferred under cold conditions (e.g. in boreal and alpine regions), because the soil N cycle is faster and N availability is higher in warmer regions than in colder regions, and (H2) soil N availability is the primary factor regulating plant N uptake preference because it provides N resources from the soil to plants.

Materials and Methods

Data collection and calculation

Peer-reviewed journal articles reporting the application of ^{15}N tracer methods to investigate plant N uptake strategies were identified by entering the following keywords in Google Scholar, the ISI Web of Knowledge, and the China Knowledge Resource Integrated Database: ‘nitrogen uptake’, ‘nitrogen preference’, ‘nitrogen strategy’, ‘nitrogen dynamics’, ‘plant nitrogen’, ‘soil nitrogen’, ‘ammonium’, ‘nitrate’, ‘organic nitrogen’, ‘ ^{15}N labeling’, ‘stable nitrogen isotope’, and ‘microbial competition’. The reference lists of relevant articles were referred to in retrieving more pertinent articles. To increase data comparability, the following criteria were applied to select appropriate observations from the initial set of articles retrieved: (1) studies containing ^{15}N uptake rates and/or the relative contribution of N forms to plant N nutrients by ^{15}N labelling method were selected, and other methods (such as ^{15}N natural abundance and concentration dilution) to explore plant N uptake strategies were excluded; (2) only the data from experiments with short isotopic exposure times were included to avoid transformations between $^{15}\text{NH}_4^+$ and $^{15}\text{NO}_3^-$, and studies that used the ^{15}N tracer method to explore long-term plant N retention, such as that of Templer *et al.* (2012), were excluded; (3) analyses restricted to species in natural plant communities in unfertilized soils were selected; (4) studies focusing on species-level N uptake but not on community-level N uptake were included; (5) as glycine (the smallest of amino acid, comprising 29% of the pool of amino-N) is a putative organic N source for plants, data using other organic N forms to represent organic N uptake were excluded from the database to make results comparable; and (6) for studies wherein uptake rates or the relative contribution of uptake rates were not directly available, we first calculated the uptake rates by dividing the ^{15}N uptake pools by the ^{15}N labelling time and root biomass. The uptake rates of each N form (NH_4^+ , NO_3^- , and glycine) were normalized to the sum of the uptake rates of the three forms to calculate their relative contributions to the total N uptake (McKane *et al.*, 2002; Xu *et al.*, 2011).

Based on these criteria, 101 publications were selected to estimate the uptake rates of different N forms (Supporting Information Notes S1). Of the 101 publications, 67 included all three N forms simultaneously. As our aim was to explore plant N

preference when NH_4^+ , NO_3^- , and organic N were considered simultaneously, we extracted 1008 observations from the 67 publications to estimate the relative contributions of the three N forms to plant N uptake. Study sites were located from $3^\circ 58' \text{S}$ to $68.63^\circ 00' \text{N}$, covering a wide range of mean annual precipitation (MAP) ($110\text{--}4600 \text{ mm yr}^{-1}$) and mean annual temperature (MAT) ($-9.6\text{--}26.6^\circ\text{C}$); the distribution of data points is shown in Fig. S1. Furthermore, as absorbing roots are mostly distributed in the topsoil, we focused on root N uptake strategies in the topsoil. The data on N uptake in the 0–10-cm soil layer accounted for more than 80% of the total dataset. Both field and pot studies were included to estimate the global average of plant N preference because the relative contributions of all three N forms did not differ significantly between the two experimental approaches (Fig. S2). However, only field studies were included in the analysis of plant N preference across climatic conditions and their relationships with the environment because the plants included in the pot studies were grown in a controlled environment and, thus, did not reflect the influence of the local climate (Liang *et al.*, 2023).

Site-specific details, such as geographical location (latitude and longitude) and climatic factors (MAT and MAP), were also extracted. If local climate data were not reported in the original paper, MAT and MAP were obtained from the WorldClim dataset (<http://www.worldclim.org/>). Ecosystem type, climate zone, and plant type were collected from the articles referred to in this study. The ecosystems were categorized as forests, grasslands, or tundra. Wetlands and heathlands were classified as ‘others’ because of several observations of these ecosystems. The climate zones were classified into (sub)tropical, temperate, boreal, alpine, and Arctic zones, as described in a previous study (Hou *et al.*, 2020). According to plant functional types, plants were categorized into nonwoody, shrub, conifer, broadleaf-deciduous, and broadleaf-evergreen species (Verheijen *et al.*, 2015). In addition, an array of soil variables was recorded, including soil physicochemical properties (bulk density, water content, and pH) and soil nutrients (organic C content, total N content, C : N ratio, NH_4^+ content, NO_3^- content, and $\text{NH}_4^+ / (\text{NH}_4^+ + \text{NO}_3^-)$ ratio). Unreported soil properties were extracted from SOILGRIDS250M 2.0 (<https://soilgrids.org>) using location information, as in another study (Wang *et al.*, 2022).

Statistical analysis

One-way analysis of variance (ANOVA) and *post hoc* analyses were performed to compare plant uptake rates and relative contributions among the three soil N forms. Differences in the N form contributions among ecosystem types, climate zones, and plant types were also analyzed using one-way ANOVA. Like in several previous studies (Wang *et al.*, 2021; Yao *et al.*, 2024), we used a linear mixed-effect model to evaluate the bivariate relationships of plant N uptake preference with variables defined as climate, soil nutrient, and soil physicochemical properties. The following equation was used: $Y = \beta_0 + \beta_1 \times X + \pi + \varepsilon$, where Y is the plant N uptake preference variable; X is the independent variable; β_0 is the intercept; β_1 is the fixed coefficient; π is the random

effect of ‘site’, accounting for the autocorrelation among observations within the site (Wang *et al.*, 2021); and ε is the sampling error. The linear mixed-effect model was performed by the maximum likelihood estimation using the LME4 package in R (Bates *et al.*, 2015). To assess the robustness of the relationships between plant N uptake preference and driving variables and clarify the relative importance of variables, a model selection analysis, based on the corrected Akaike information criterion, was used. Specifically, the importance of each variable as the sum of the Akaike weights was calculated using the GLMULTI package, and a cutoff of 0.8 was set to distinguish between the important and unimportant variables (Calcagno & Mazancourt, 2010), as in several previous studies (Cheng *et al.*, 2022; Zhou *et al.*, 2022). In order to verify the robustness of the results, a multivariate linear mixed model and hierarchical partitioning analysis were also used to assess the importance of factors to predict plant N uptake preference (Jin *et al.*, 2024; Teng *et al.*, 2024).

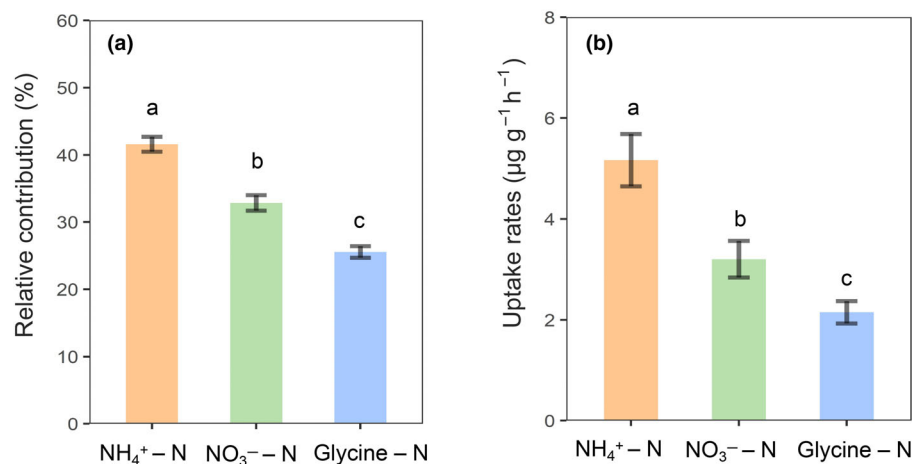
To quantify the direct and indirect effects of multiple driving variables on plant N preferences, a structural equation model (SEM) was then used. This approach can partition the direct and indirect effects that one variable may have on another and is therefore useful for exploring complex relationships in natural ecosystems (Grace, 2006; Chen *et al.*, 2019). SEM testing was performed using the PIECEWISESEM and LME4 packages in R. Directional-separation tests were applied to check for any missing significant paths and for unnecessary insignificant paths that needed to be excluded. The final model was selected using the Akaike information criterion and P -values ($P > 0.05$ was considered adequate). Two SEMs were independently run using the same procedure described above to quantify the effects of multiple variables on plant NH_4^+ and NO_3^- uptake preferences. The SEM of organic N was not run because there were no significant relationships between glycine preference and most of the factors. Data were tested for normality and log-transformed when required before statistical analyses. All analyses were performed using the R software (v.4.1.0, R Core Team, 2020), and significance was set at $P < 0.05$.

Results

Global patterns of plant N uptake preference

The relative contribution of soil NH_4^+ to plant N uptake ($41.6 \pm 1.1\%$) was significantly higher than that of NO_3^- ($32.8 \pm 1.2\%$) and glycine N ($25.6 \pm 0.9\%$) (Fig. 1a). Correspondingly, the average uptake rate of NH_4^+ was significantly higher than that of NO_3^- and glycine N (Fig. 1b). Plant N uptake preferences varied significantly with the climate zone and biome type (Fig. 2a–f). The contribution of NH_4^+ was significantly higher in (sub)tropical regions ($65.6 \pm 1.8\%$) than in other climate zones ($39.4 \pm 2.4\%$, $44.4 \pm 3.3\%$, $32.3 \pm 1.3\%$, and $39.4 \pm 3.2\%$ in temperate, boreal, alpine, and Arctic climate zones, respectively) (Fig. 2a). Plant NO_3^- preference was highest in alpine regions ($42.1 \pm 1.7\%$) and lowest in (sub)tropical regions ($15.4 \pm 1.7\%$). Plant NO_3^- preference showed similar levels between Arctic ($36.4 \pm 3.9\%$), boreal

Fig. 1 Relative contribution (%) of soil nitrogen (N) forms (NH_4^+ , NO_3^- , and glycine N) (mean $\pm$ SE) to plant N uptakes (a) and uptake rates of N forms (b) at the global scale. The relative contribution was directly reported or indirectly calculated by dividing the uptake rates of each N form by the sum of the uptake rates of the three forms. Different letters on the top of error bars represent significant differences among soil N forms ($P < 0.05$, one-way analysis of variance with *post-hoc* test).



($34.3 \pm 3.6\%$), and temperate regions ($32.4 \pm 2.9\%$) (Fig. 2b). Furthermore, plant preference for glycine N was lowest in (sub) tropical regions ($19.1 \pm 1.6\%$), which was significantly lower than that in temperate and alpine regions (Fig. 2c).

Regarding biome types, the preference for NH_4^+ was higher in forests ($53.6 \pm 2.1\%$) than in grasslands ($32.2 \pm 1.4\%$) and tundra ecosystems ($34.9 \pm 3.1\%$) (Fig. 2d). By contrast, the preference for NO_3^- was higher in grasslands ($42.5 \pm 1.9\%$) and tundra ($42.2 \pm 3.8\%$) than in forests ($21.2 \pm 2.0\%$) (Fig. 2e). Meanwhile, the preference for organic N did not differ significantly between biome types (Fig. 2f).

Changes in plant N uptake preference across different plant functional groups

Plant N uptake preference varied significantly between different plant functional groups according to the statistical analysis results of one-way ANOVA (Fig. 3). Broadleaf-evergreen species had the highest preference for NH_4^+ , while nonwoody species had the lowest preference for NH_4^+ (Fig. 3a). Conversely, the preference of nonwoody species for NO_3^- was significantly higher than that of other plant functional types (broadleaf-evergreen, conifer, and shrub species) (Fig. 3b). Preference for organic N did not differ significantly between plant functional types (Fig. 3c).

Effects of climate, soil, and plant functional group on plant N uptake preference

Plant preference for NH_4^+ and NO_3^- uptake was significantly correlated with climatic factors and soil properties (Fig. 4). The preference for NH_4^+ was positively correlated with MAT (Fig. 4a), MAP (Fig. 4d), $\text{NH}_4^+ / (\text{NH}_4^+ + \text{NO}_3^-)$ ratio (Fig. 4g), and C : N ratio (Fig. 4j), but negatively correlated with soil pH (Fig. 4m). By contrast, the preference for NO_3^- increased significantly with soil pH (Fig. 4n); however, it decreased with increasing MAT (Fig. 4b), $\text{NH}_4^+ / (\text{NH}_4^+ + \text{NO}_3^-)$ ratio (Fig. 4h), C : N ratio (Fig. 4k), and soil moisture (Fig. 4q).

Model selection analysis showed that among the multiple abiotic and biotic (plant functional type) factors, MAT and

$\text{NH}_4^+ / (\text{NH}_4^+ + \text{NO}_3^-)$ ratio were the most important drivers of NH_4^+ preference (Fig. 5a,c), whereas MAT, soil water content, $\text{NH}_4^+ / (\text{NH}_4^+ + \text{NO}_3^-)$ ratio, and plant functional type were the main drivers of NO_3^- preference (Fig. 5b,d). Consistently, the results of the multivariate linear mixed model also showed that MAT and soil $\text{NH}_4^+ / (\text{NH}_4^+ + \text{NO}_3^-)$ were the two main determinants of plant NH_4^+ uptake preference, while both environmental factors and plant functional types were significant drivers of plant NO_3^- uptake preference (Fig. S3). Plant preference for organic N uptake showed insignificant relationships with most of the factors (Figs 4, S4).

Discussion

Our study showed that, on a global scale, soil $\text{NH}_4^+ - \text{N}$ is the major contributor to plant N uptake, followed by $\text{NO}_3^- - \text{N}$, although organic N is also an important N resource for plants. The considerable variation in plant NH_4^+ and NO_3^- preference between different regions and biomes (Fig. 2) is mainly due to their different climate and soil properties, as well as different plant functional groups (Fig. 3). These results enrich the current knowledge on plant N uptake strategies and ecosystem N cycling. Most land models (such as CABLE and CLM4) simulate plant N acquisition based only on plant N demand and soil N availability without considering the effects of plant N preference strategies (Thornton *et al.*, 2007; Wang *et al.*, 2010; Zaehle *et al.*, 2014). Based on our findings, we recommend incorporating plant N preference into C–N models to better assess N cycling and its impact on land C balance.

Different plant N uptake preference among climate zones

Our first hypothesis was supported by the significant differences in preferences of plants for different soil N sources, reinforcing the idea that natural selection causes species to diverge into highly specialized niches for N consumption (McKane *et al.*, 2002; Moreau *et al.*, 2019). Plants usually absorb less NO_3^- than NH_4^+ because of the additional C required for NO_3^- uptake (Britto & Kronzucker, 2013; Zou *et al.*, 2020). However, our

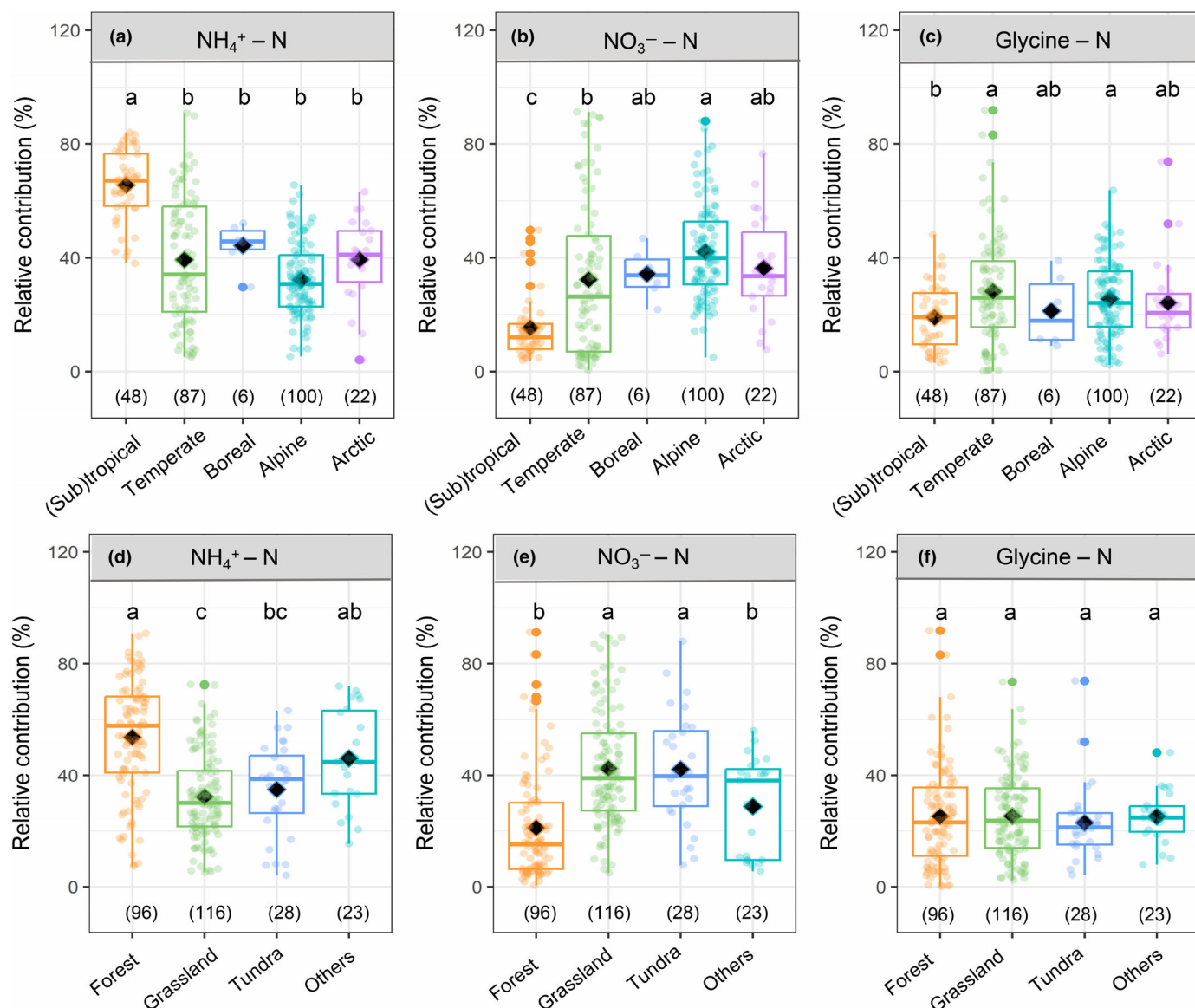


Fig. 2 Relative contribution of different soil nitrogen (N) forms (NH_4^+ , NO_3^- , and glycine N) to plant N uptake varied with climate zones (i.e. subtropical, temperate, boreal, alpine, and arctic) (a–c) and biome types (i.e. forest, grassland, tundra, and other ecosystems) (d–f). Box plots indicate the medians (horizontal lines), first and third quartiles (boxes). The black diamond in each box is the mean value, and jittered points represent individual observations. Values in parentheses represent sample sizes, which are the same for different N forms because only data from studies that used three different N forms simultaneously were analyzed. Different lowercase letters at the top of the boxes indicate significant differences among respective groups ($P < 0.05$, one-way analysis of variance with *post-hoc* test).

results are inconsistent with a recent study that found a slightly higher contribution of NO_3^- to plant N uptake than that of NH_4^+ (Liu *et al.*, 2024). Unlike our findings, which focus on natural ecosystems without human interference (e.g. N fertilizer addition), the results of Liu *et al.* (2024) include data from arable soils. In intensively managed agricultural soils, mineral N and, especially, NO_3^- are the most abundant forms of available N (Wirén *et al.*, 1997). NO_3^- is the predominant form of N taken up by most crops, such as maize (*Zea mays* L.), wheat (*Triticum aestivum* L.) (Crawford & Glass, 1998; Gao *et al.*, 2022). This may explain the inconsistent results between the two studies.

Meanwhile, we found that plant N preference for soil NH_4^+ and NO_3^- varied across climate zones (Fig. 2). The significantly higher plant preference for NH_4^+ in (sub)tropical zones than in boreal, alpine, and Arctic regions is consistent with previous theories that warm conditions are favorable for plant NH_4^+ uptake (Houlton *et al.*, 2007; Liu *et al.*, 2017). The varying plant preference for NH_4^+ among different climatic regions may be partly due to their differential N availability. Decomposition (more NH_4^+ availability) and leaching rates of NO_3^- are higher in (sub)tropical regions than in high-latitude regions, leading to a higher NH_4^+ relative availability. This provides plants more

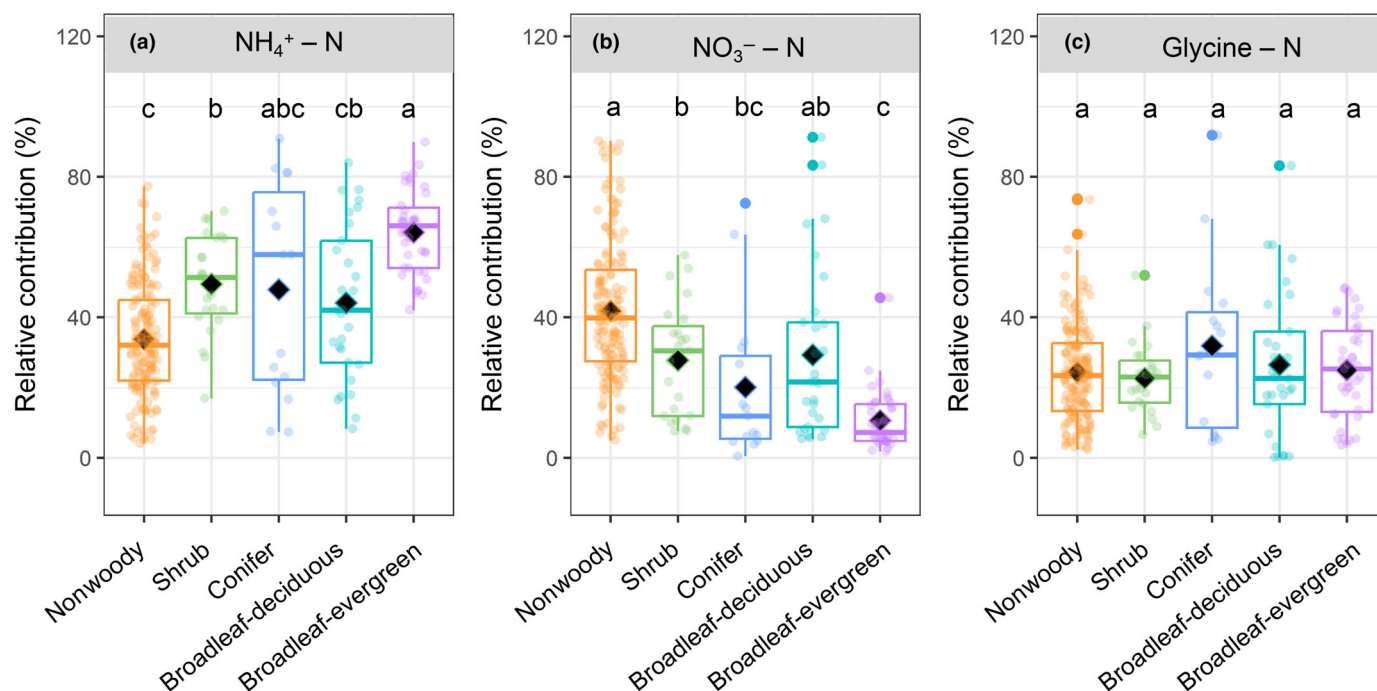


Fig. 3 Relative contribution of different soil N forms (NH_4^+ , a; NO_3^- , b; and glycine N, c) to plant N uptake varied with plant functional types. Box plots indicate the medians (horizontal lines), first and third quartiles (boxes). The black diamond in each box is the mean value, and jittered points represent individual observations. Different lowercase letters at the top of the boxes indicate significant differences among respective groups ($P < 0.05$, one-way analysis of variance with *post-hoc* test).

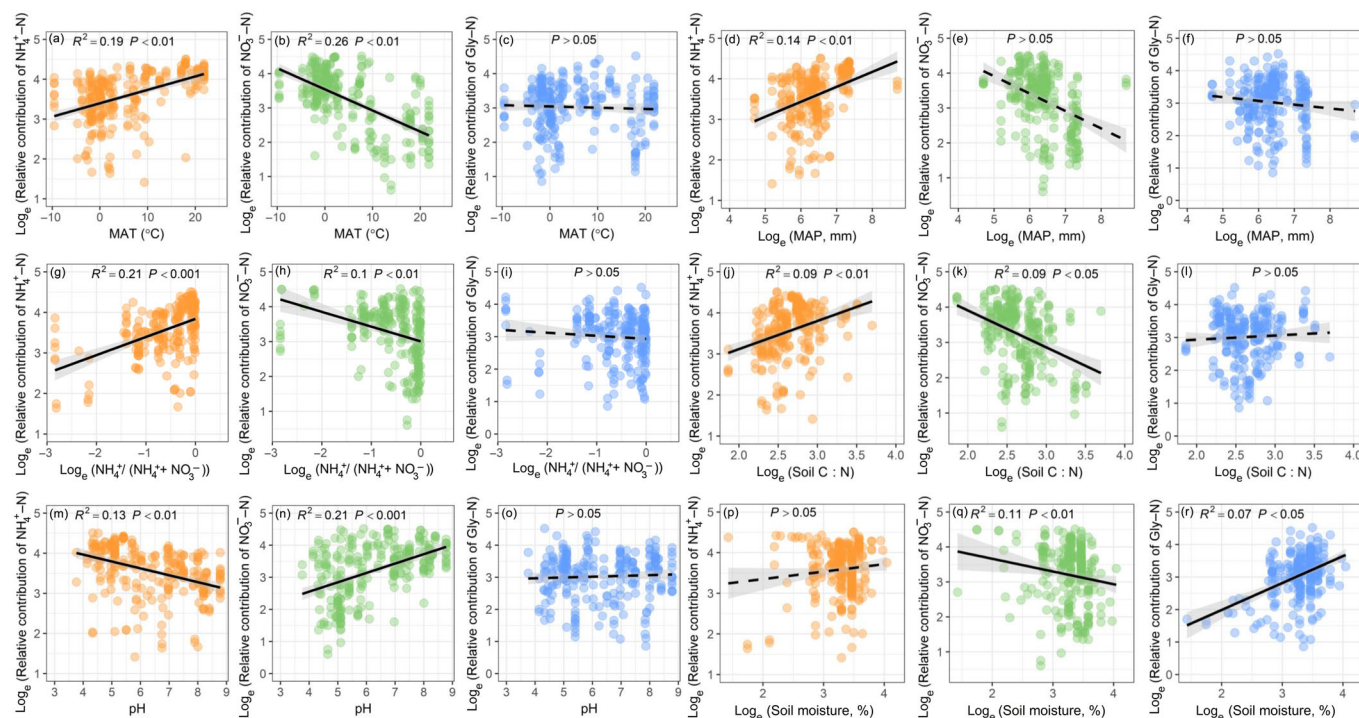


Fig. 4 Contribution of soil NH_4^+ -N, NO_3^- -N, and glycine N to plant nitrogen (N) uptake in relation to mean annual temperature (MAT, a–c), mean annual precipitation (MAP, d–f), soil $\text{NH}_4^+ / (\text{NH}_4^+ + \text{NO}_3^-)$ ratio (g–i), soil C : N ratio (j–l), pH (m–o), and soil moisture (p–r) at the global scale. Regression lines and shaded areas depict the best-fit trendline and the 95% confidence interval of the linear regression, respectively. Solid lines represent significant correlations at $P < 0.05$ based on the maximum likelihood test, and dashed lines represent insignificant correlations at $P > 0.05$.

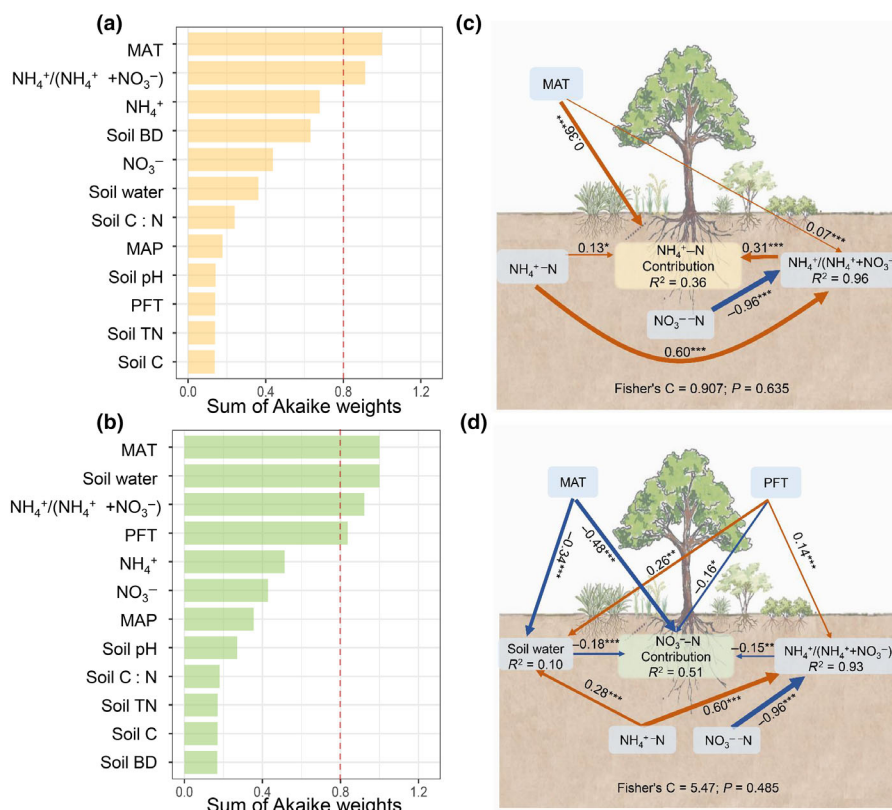


Fig. 5 Model-averaged importance of potential predictor variables for plant NH_4^+ (a) and NO_3^- (b) uptake preference based on the sum of Akaike weights derived from the model selection analysis. Structural equation modelling (SEM) showing the multiple relationships of NH_4^+ (c) and NO_3^- (d) preference with driving factors at the global scale. In (c, d), the values are the standardized path coefficients from the model. The numbers beside the arrows are standardized path coefficients. The thickness of the arrows is proportional to the magnitude of the standardized path coefficients. R^2 stands for the amount of variation interpreted by variables, and the significant level is set at *, $P < 0.05$; **, $P < 0.01$; ***, $P < 0.001$ based on structural equation model analysis. MAP, mean annual precipitation; MAT, mean annual temperature; $\text{NH}_4^+ / (\text{NH}_4^+ + \text{NO}_3^-)$, the ratio of soil ammonium N to ammonium plus nitrate N; soil C : N, the ratio between soil organic carbon and total nitrogen. Soil BD, soil bulk density; Soil water, soil water content; Soil pH, soil pH values; PFT, plant functional types. The glycine preference (organic N) was not included in the figure due to its insensitivity to most environmental factors.

opportunities for NH_4^+ uptake in (sub)tropical regions (Britto & Kronzucker, 2013; Zou *et al.*, 2020). Furthermore, high temperatures in (sub)tropical regions may lead to high diffusive fluxes of soil NH_4^+ , which may contribute to a higher NH_4^+ preference in (sub)tropical regions than in other climate zones, as NH_4^+ has extremely low mobility in the soil (County *et al.*, 2015).

Plant NO_3^- preference was higher in the alpine and Arctic zones than in lower latitudes (Fig. 2). This result is consistent with some previous studies (Choudhary *et al.*, 2016; Liu *et al.*, 2018) that emphasize the important role of soil NO_3^- in plant N uptake in the Arctic tundra, but in contrast to other studies that reported a low soil nitrification rate associated with undetectable soil NO_3^- , which is unfavorable for plant NO_3^- uptake (McKane *et al.*, 2002; Nordin *et al.*, 2004). These conflicting results suggest that factors other than soil NO_3^- availability play a dominant role in driving plant NO_3^- preference. We propose the following reasons for the high NO_3^- preference in cold regions. First, compared with NH_4^+ , negatively charged

NO_3^- adheres poorly to the negatively charged surfaces of clay minerals and soil organic matter. Hence, being more mobile than NH_4^+ , NO_3^- can more easily move to the root surface, increasing the likelihood of its uptake (Zhou *et al.*, 2021), especially in high-latitude ecosystems, where soil N availability is limited (Du *et al.*, 2020; Mason *et al.*, 2022). By contrast, in (sub)tropical regions, where the amount of rainfall is usually high (Adler *et al.*, 2017), soil NO_3^- is easily lost through leaching, which may result in a low NO_3^- contribution to plant N uptake. In line with this, our results highlighted the significant effect of soil water content on plant NO_3^- uptake preference (Fig. 5), providing robust evidence that soil moisture plays an important role in plant NO_3^- uptake preference (Houlton *et al.*, 2007). Second, in the present study, most plants in high-latitude ecosystems (alpine and Arctic) were nonwoody species with a significantly higher proportion of NO_3^- uptake (Fig. 3), which may have contributed to the high NO_3^- preference at high latitudes, as plant functional types played an important role in regulating the variations of plant NO_3^- preference (Figs 5, S3). The variation in NO_3^-

preference between plant functional types may be due to the unique functions and life strategies of different plant species. For example, previous evidence has shown that NO_3^- is much easier to store in plant tissues than NH_4^+ (Boudsocq *et al.*, 2012; Hachiya & Sakakibara, 2016). Nonwoody plants are less susceptible to pathogens than woody plants (Jiang *et al.*, 2024). A higher NO_3^- uptake rate may allow nonwoody species to invest more N in defense-related secondary metabolites. These results suggest that further studies on different plant functional groups across broad climatic regions are warranted to further verify the climatic patterns observed in the current study.

It has been suggested that compared with low-latitude ecosystems, plants in high-latitude ecosystems rely more on soil dissolved organic N, where much of the N pool is bound to organic matter and must be depolymerized by soil microorganisms whose activity is limited by low temperatures (Näsholm *et al.*, 2009; Britto & Kronzucker, 2013; Hu *et al.*, 2024). In partial support of this hypothesis, we found that compared with other climatic regions, the preference for glycine was lowest in (sub)tropical regions (Fig. 2), which typically have abundant soil inorganic N sources (Du *et al.*, 2020; Mason *et al.*, 2022). However, there were insignificant differences in organic N preference between subtropical and boreal or arctic regions. Similarly, no significant changes in organic N uptake were also observed among biomes or plant functional types. The reason for the patterns of plant organic absorption may be partly due to the effect of soil water content. Soil moisture can significantly change plant glycine N absorption by affecting glycine diffusion and migration from the soil to plant roots (McMurtrie & Näsholm, 2018; Liang *et al.*, 2022). In the current study, we found that among multiple environmental factors, soil water content was the most important determinant of plant preference for organic N uptake (Figs 4, S5a). However, there were no significant differences in soil water content among biomes or plant functional types (Fig. S5b,c), which may result in insignificant changes in organic N uptake among biomes or plant functional types. It is worth noting that the significance and explanatory power of the current environmental variables for organic N are low (Fig. 4). This suggests that there are other factors (e.g. soil microbial competition, plant physiology, and mycorrhizal symbiosis) (Näsholm *et al.*, 2009; Tünnermann *et al.*, 2025) that regulate the patterns of organic N uptake. Unfortunately, these factors could not be included in the current analysis due to a lack of data in the literature.

Influences of climate factors and soil properties on plant N uptake preference

Temperature can significantly affect plant uptake of different N forms by changing root transport properties for different N forms (Tegeder & Masclaux-Daubresse, 2017; Lee *et al.*, 2024) or by mediating ion movement through the roots (Bassirirad, 2000; Kuster *et al.*, 2016; Sanders-DeMott *et al.*, 2018). An increase in temperature can increase N diffusion in soil because solute diffusion rates in a liquid medium are proportionally related to temperature (Kelly & Mays, 1999; Inselsbacher & Näsholm, 2012). There is evidence that diffusive fluxes of NH_4^+ in soils increase

significantly with increasing soil temperature (Inselsbacher & Näsholm, 2012), thereby enhancing plant uptake of NH_4^+ (Bassirirad, 2000; Warren, 2009). Furthermore, warming stimulates microbial activity, leading to accelerated N mineralization and increased soil NH_4^+ availability (Risch *et al.*, 2019), which may explain the positive correlations between MAT and NH_4^+ preference (Figs 4, 5). By contrast, as NO_3^- moves easily through the soil solution, temperature-induced diffusion and mass flow increase the risk of NO_3^- loss, resulting in a negative correlation between MAT and NO_3^- preference (Figs 4, 5). Overall, the significantly different correlations of MAT on plant NH_4^+ and NO_3^- uptake preferences suggest that species composition may change under global warming due to the significant difference in N nutrient and growth between plant species. However, to date, only a few experimental studies have been conducted to determine how warming may affect plant N uptake strategies (Boczulak *et al.*, 2014; Kuster *et al.*, 2016). Therefore, more empirical studies are needed to better understand plant N uptake strategies and N-induced plant C storage under global warming.

Although the regulation of plant N preference is commonly attributed to soil N availability (Nordin *et al.*, 2004; Britto & Kronzucker, 2013), we found that the proportions of N forms in the soil also regulate plant NH_4^+ or NO_3^- preference (Figs 4, 5). The importance of the relative availability of soil N forms in driving plant preference for NH_4^+ or NO_3^- extends our traditional understanding of plant N uptake strategies, which mainly depend on soil N content (Schimel & Bennett, 2004; Niu *et al.*, 2016). This finding suggests that we cannot judge the utilization ability and fractional contribution of a given N resource to plants based on soil N availability alone. A number of recent studies have revealed significant spatial variability in the shifts in N deposition composition (i.e. NH_4/NO_3) (Li *et al.*, 2016; Yu *et al.*, 2019), which may lead to changes in soil N form availability and distinct fates of plants with different N preferences and associated ecosystem functions, particularly plant C sinks.

Soil water availability is also a crucial factor driving plant N uptake due to nitrogen–water interactions (Wang *et al.*, 2013; Wu *et al.*, 2022). It has been noted that plants prefer NO_3^- uptake under dry soil conditions, whereas plants prefer NH_4^+ uptake in wetter soil environments (Houlton *et al.*, 2007; Wang & Macko, 2011). Our results support those of previous studies showing a significant negative correlation between soil water content and plant NO_3^- uptake preference, which may be partly due to the high mobility of soil NO_3^- . The higher mobility of NO_3^- in the soil and its faster transport within plants enhance NO_3^- diffusion to the root surface and transport to aboveground plant parts (County *et al.*, 2015), resulting in a higher NO_3^- uptake preference under dry conditions. However, under wet conditions, there is a high risk of leaching losses of NO_3^- . Therefore, soil water content negatively affects plant NO_3^- preference.

Study limitations and implications for future research

This synthesis revealed global patterns and drivers of plant N acquisition preference across various ecosystems and climatic regions, but further studies are needed to fully understand these

aspects. First, the uneven distribution of study sites across different climate zones in the current study may lead to biases in the average plant N preference (e.g. overestimation of glycine N uptake preference due to significantly more data in alpine and temperate regions with greater organic N preference than that in subtropical regions) on a global scale (Persson & Näsholm, 2008). In addition, the pattern of plant preference for organic N uptake in our study was based on glycine. Recent studies have suggested that plants can take up different organic N forms besides amino acids (Liu *et al.*, 2017; Moreau *et al.*, 2019). Therefore, the conclusions drawn from a simple glycine model may not fully reflect the potential organic N uptake by plants (Warren, 2013). To accurately assess the ecological significance and importance of organic N as a plant N resource, further experiments using multiple organic N forms labelled with ^{13}C – ^{15}N in broader climatic regions are recommended. Additionally, it is unclear whether plant N preference strategies and the factors regulating such strategies are consistent between the species and ecosystem levels. More field observations of N preference at community and ecosystem scales are needed to further extend our findings.

Second, although we have done our best to include factors to clarify the drivers influencing the variations of plant N uptake preference, a significant proportion of the variability in plant N preference remains unexplained, in part due to the limited data on soil microorganisms. Soil microorganisms strongly control plant N uptake (Hill & Jones, 2019). In the short term, microbes are more efficient than plants at inorganic and organic N uptake (Harrison *et al.*, 2008). This may be due to the rapid growth rates and high surface-area-to-volume ratios of microorganisms compared with plant roots in different ecosystems (Kuz'yakov & Xu, 2013). Experiments have shown that plants can compete with soil microbes for amino acids as well as for NH_4^+ (Mckane *et al.*, 2002), contrary to the previous speculation. In this study, soil microbial biomass N showed insignificant correlations with plant N uptake preference (Fig. S6), which may be partly due to limited data (less than one fifth of the studies included information on microbial biomass). To fully understand how plants take up N, it is important to understand how N interacts with microorganisms. Therefore, further studies using ^{15}N tracers are needed to compare how plants take up N and how microorganisms immobilize it at the same time. Furthermore, although we have considered the role of plant functional types on plant N preference, the role of phylogeny in driving plant N preference is still an open question. To better understand plant N preference strategy and underlying mechanisms under environmental change, further investigation of plant N preference responses in a phylogenetic context is needed.

In summary, this study is one of the first to investigate the global patterns and driving factors of the relative importance of different soil N forms for plant N uptake using a ^{15}N tracer. Our results showed clear climatic patterns of plant preference for soil NH_4^+ and NO_3^- and low preference for organic N in (sub)tropical zones. Furthermore, we found that different factors determine plant N preference strategies for different soil N forms. Plant NH_4^+ uptake preference was mainly regulated by MAT and soil $\text{NH}_4^+ / (\text{NH}_4^+ + \text{NO}_3^-)$ ratio, whereas NO_3^- uptake preference was mainly driven by both environmental factors (MAT, soil

water content, $\text{NH}_4^+ / (\text{NH}_4^+ + \text{NO}_3^-)$) and plant functional types. These results extend our traditional understanding of plant N uptake strategies, which mainly focus on soil N richness. As the atmospheric deposition of NH_4^+ -N and NO_3^- -N is uneven in most areas and climate warming continues to increase, it is critical to clearly identify plant N preference to predict N limitation to C sequestration in the face of increasing global change.

Acknowledgements

We thank the anonymous reviewers for their valuable comments and suggestions on the manuscript. This work was financially supported by the National Natural Science Foundation of China (31988102, 32301394, and 32241035), the National Key R&D Program of China (2022YFF0802102), and the China Postdoctoral Science Foundation (2023M743457).

Competing interests

None declared.

Author contributions

SN and JM conceptualized the research plan. JM collected and analyzed the data and wrote the first manuscript. SN and JW heavily revised the manuscript with inputs from DT, RZ and JP. JL and XX provided great help during data collection. SN, JM, JW, JL, XX, DT, RZ and JP discussed the article. All authors contributed to the reviewing of the manuscript.

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Data availability

The data that support the findings of plant N preferences are available in the Supporting Information of this article (Table S1).

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Supporting Information

Additional Supporting Information may be found online in the Supporting Information section at the end of the article.

Fig. S1 The distribution of experimental sites in this study.

Fig. S2 Comparisons for the plant N uptake preference between *in situ* and pot experiments.

Fig. S3 Parameter estimates of the model predictors for plant NH_4^+ and NO_3^- uptake preference.

Fig. S4 Relationship of the relative contribution of soil NH_4^+ , NO_3^- and glycine N to plant total N uptake with soil organic C, total nitrogen, and bulk density at the global scale.

Fig. S5 Model-averaged importance of potential predictor variables for plant organic N uptake preference (a). The differences in soil water content between biomes (b) and plant functional types (c).

Fig. S6 Relationship of the relative contribution of soil NH_4^+ , NO_3^- and glycine N to plant total N uptake with soil microbial biomass N at the global scale.

Notes S1 data sources.

Table S1 Data file including the variables and references related to plant nitrogen uptake preference.

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